Exercise on Seating Arrangements

- In how many ways, can 5 boys and 3 girls sit around a circular table so that no two girls are together? (Order matters. Since people sit on a circular table, the arrangement 12 345 678 and 23 456 781 are considered to be same.)
- Solution
- There are 4! seating arrangements for the 5 boys, when rotations are equivalent (see slide 11).
- For each girl not to seat next to another girl, she must have a boy to the left in C(5,3) ways (also a boy to the right). Moreover, there are 3! permutations for the girls. Thus, C(5,3)4!3!